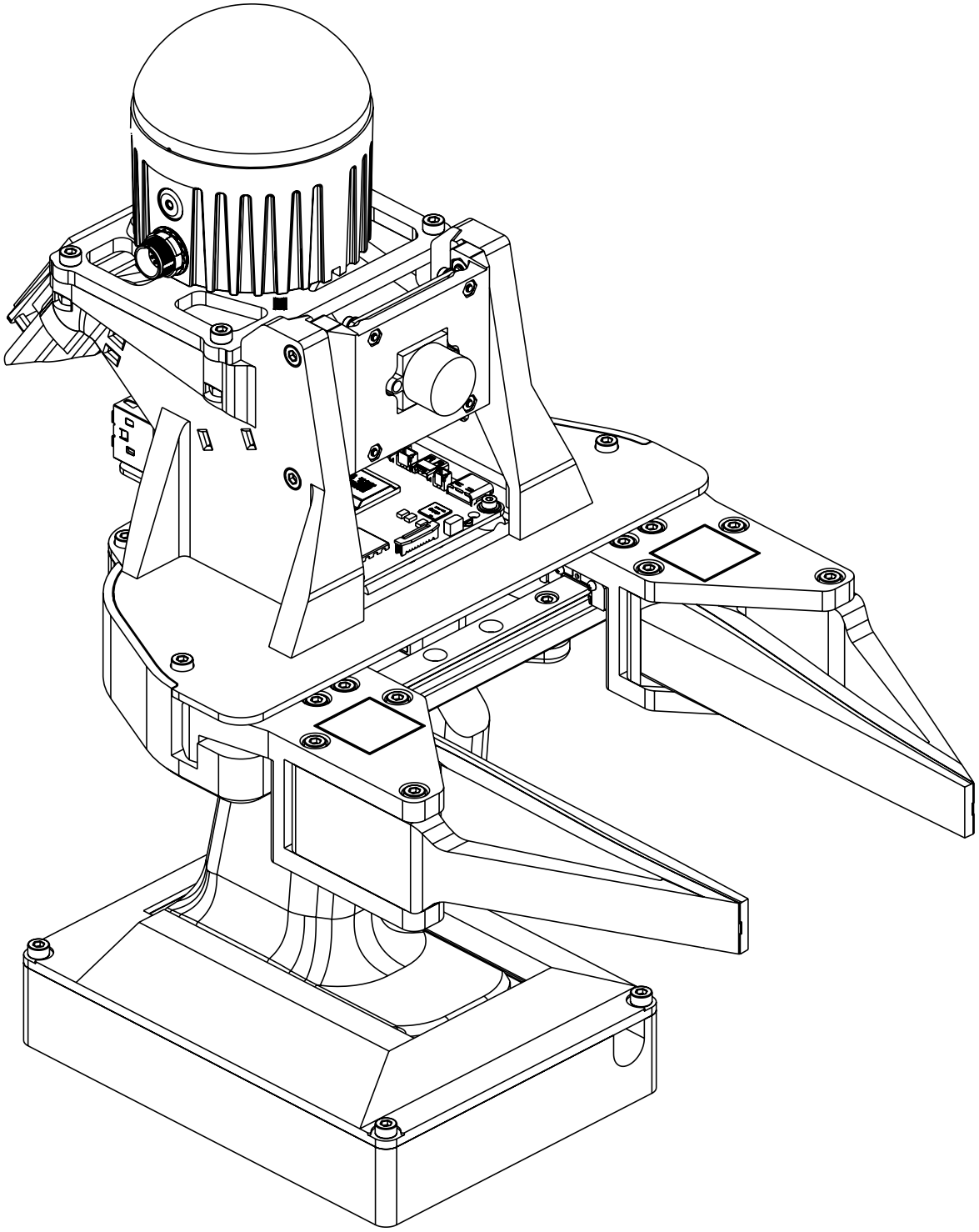
UMI-Live

Hardware Instructions



ASSEMBLY GUIDE

MECHANICAL / ELECTRONICS / WIRING

v6

Table of Contents

SECTION	PAGE
Parts and hardware	3 - 5
Body, trigger and gears	6 - 11
Fingers and ArUco tags	11 - 15
Body top and Pi	16 - 17
LCD and button pin map	18
Pi, camera, LCD and LiDAR	19 - 22
Battery pack	23 - 26
Connect and inspect	27

Notes

This guide assembles one UMI-Live gripper. Build two units for a bimanual setup.

Recommended Tools: M2/M3 drivers, scissors, flush cutters, multimeter, wire stripper.

3D Prints: remove and clean all support materials from printed parts.

Carriages: select two carriages that slide the smoothest on the rail.

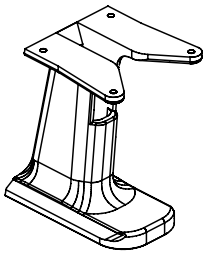
> **!** Keep carriages on the rail at all times to prevent the ball bearings from falling out.

Threadlocker: optional for long-term or heavy use on metal-to-metal threads.
Keep it off plastic, bearings and sliding surfaces.

> **!** Keep the battery disconnected while assembling or changing wiring.

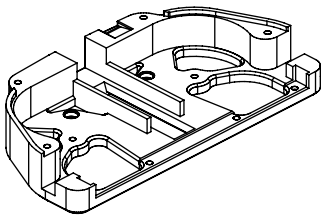
3D printed parts

grip



A **1x**

body_bottom



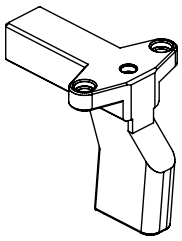
B **1x**

gear_axle



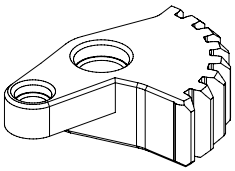
C **2x**

trigger



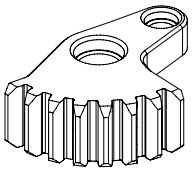
D **1x**

gear_left



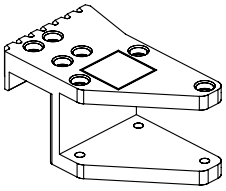
F **1x**

gear_right



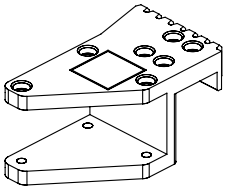
G **1x**

finger_holder_left



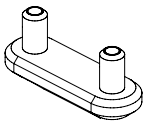
H-L **1x**

finger_holder_right



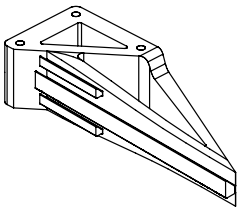
H-R **1x**

linkage



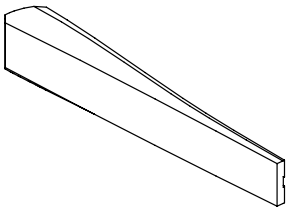
E **2x**

finger_backbone



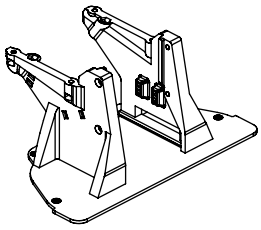
I **2x**

finger_pad



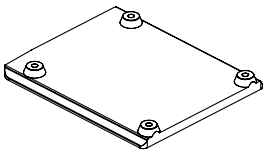
J **2x**

body_top



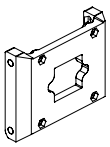
K **1x**

raspberry_slider



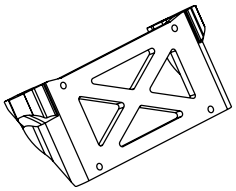
L **1x**

camera_mount



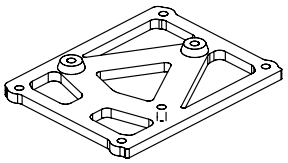
M **1x**

lcd_mount



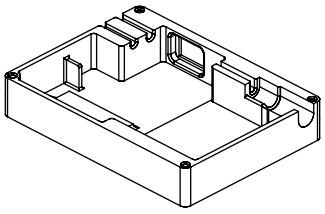
N **1x**

lidar_mount



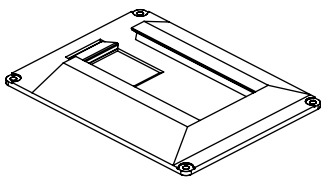
O **1x**

battery_pack_bottom



P **1x**

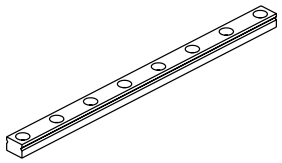
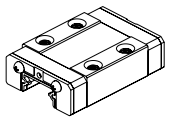
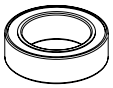
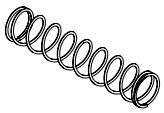
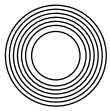
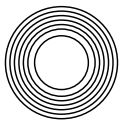
battery_pack_top



Q **1x**

Illustrations scaled for identification; small parts are enlarged.

Mechanical parts

MGN9 150mm Linear Rail  R **1x**	**MGN9C Carriage Blocks**  S **2x**	**uxcell MR85-2RS Ball Bearings**  T **8x**	**uxcell MR128-2RS Ball Bearings**  U **4x**
uxcell Compression Spring, 10x0.7x50mm  V **1x**	**ArUco tags (DICT_4X4_50, 16 mm)**  AM **2x**	**3M Gripping Material TB641, Black**  AN **Cut to fit**	**UHMW Polyethylene Transparent Film Slick Tape**  AP **Cut to fit**

Fasteners

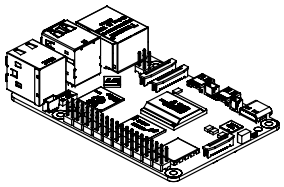
	H1 **M3 × 8 mm**	**14x**
	H2 **M3 × 12 mm**	**15x**
	H3 **M3 × 16 mm**	**2x**
	H4 **M3 × 20 mm**	**2x**
	H5 **M3 × 25 mm**	**10x**
	H6 **M3 × 30 mm**	**1x**
	H7 **M3 × 35 mm**	**6x**

	H8 **M3 nut**	**34x**
	H9 **M3 washer**	**2x**
	H10 **M2 × 6 mm**	**4x**
	H11 **M2 × 8 mm**	**8x**
	H12 **M2 nut**	**8x**
	H13 **M2 washer**	**12x**

Illustrations scaled for identification; small parts are enlarged.

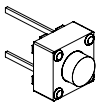
Electronics and cables

Raspberry Pi 5



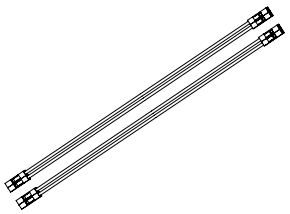
W **1x**

Tactile Button 2 Pin



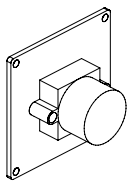
X **2x**

10cm Breadboard Jumper Wires Assorted Kit



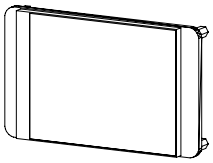
AL **4x**

SVPRO 2MP Global Shutter USB Camera



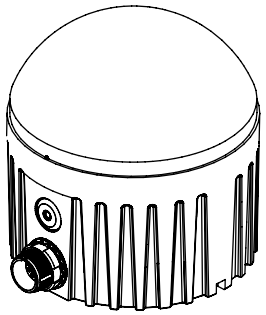
Y **1x**

Waveshare 2inch LCD Display SPI / 240 x 320



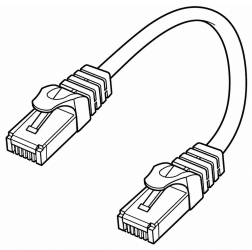
Z **1x**

RoboSense Airy LiDAR



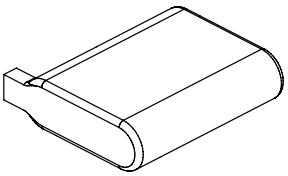
AA **1x**

Short Ethernet Cable 0.5FT



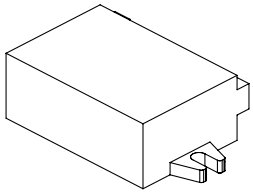
AK **1x**

12V 2600mAh Rechargeable Li-ion Battery Pack



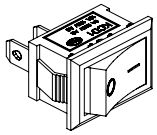
AB **1x**

DC-DC 12V/24V to 5V 5A 25W Buck Converter USB-C output



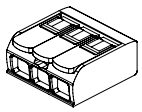
AC **1x**

DaierTek Mini ON and Off Rocker Switch



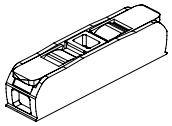
AD **1x**

WAGO 221-413 Lever Nuts



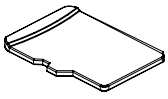
AF **2x**

WAGO 221-2401 Lever Nuts



AG **1x**

microSD card

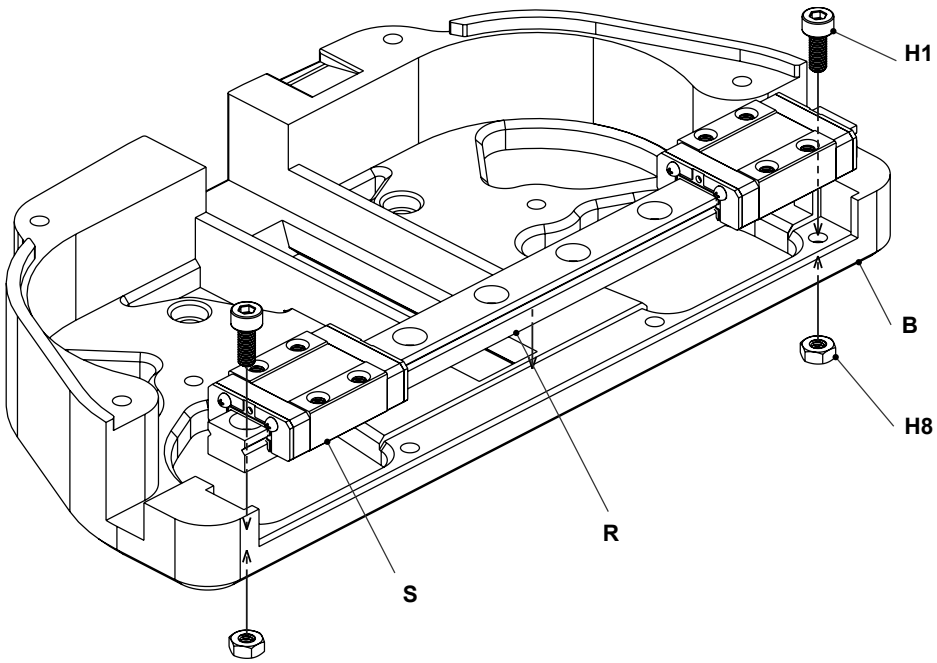


AH **1x**

Illustrations scaled for identification; small parts are enlarged.

STEP 1

Fit the rail and carriages



Select the two smoothest carriage blocks. Secure the rail to the body bottom using two M3 x 8 screws in the outer holes and two nuts underneath. Leave the two inner holes empty for Step 2.

Keep both carriages on the rail; do not run them off an end.

PARTS ADDED

B **1x**
Body bottom

R **1x**
Linear rail

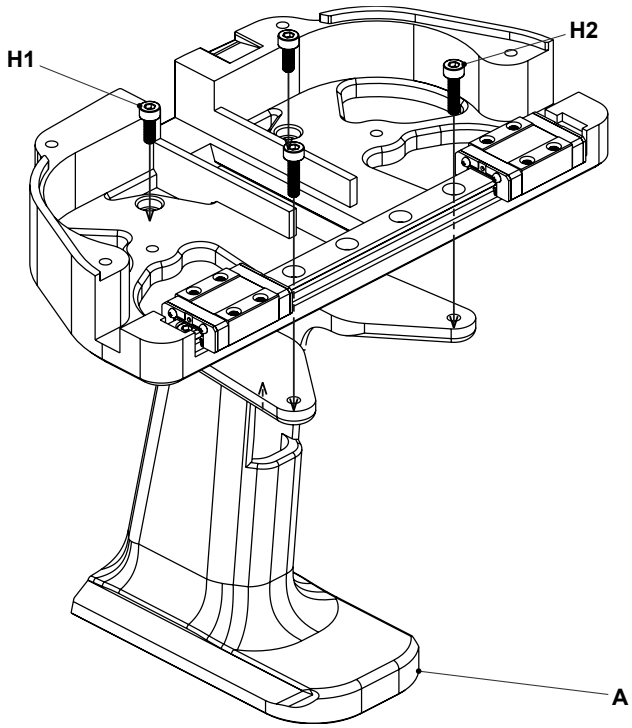
S **2x**
Carriage block

H1 **2x**
M3 × 8 mm

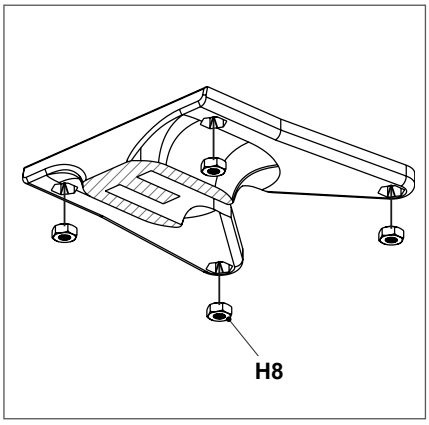
H8 **2x**
M3 nut

STEP 2

Attach the grip



GRIP FLANGE / UNDERSIDE CUT



Align the grip beneath the body bottom. Fit two M3 x 12 screws through the inner rail holes, body bottom and grip, plus two M3 x 8 screws through the other grip mounting holes. Secure all four screws with nuts underneath and tighten evenly.

PARTS ADDED

A **1x**
Grip

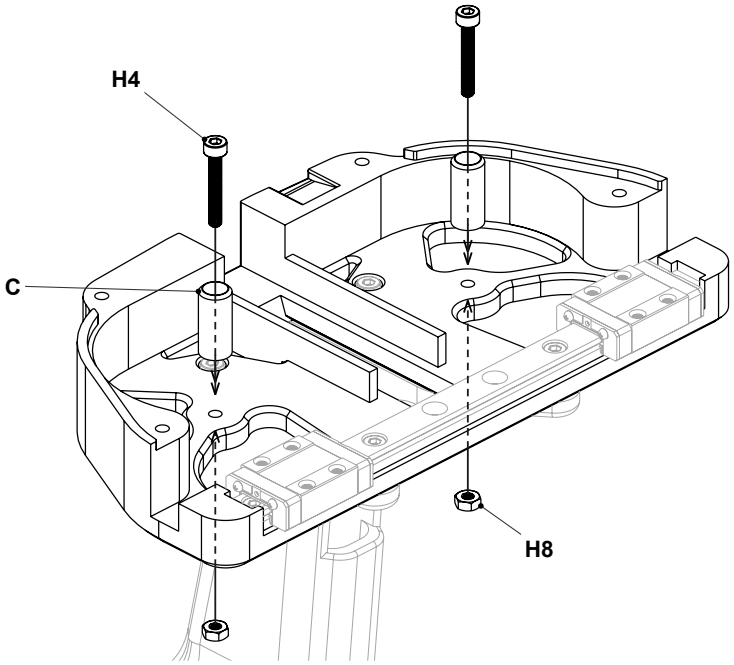
H1 **2x**
M3 × 8 mm

H2 **2x**
M3 × 12 mm

H8 **4x**
M3 nut

STEP 3

Install the two gear axles



Seat one printed gear axle at each pivot location. Secure each axle with one M3 x 20 screw and nut. Keep the axles upright and fully seated.

PARTS ADDED

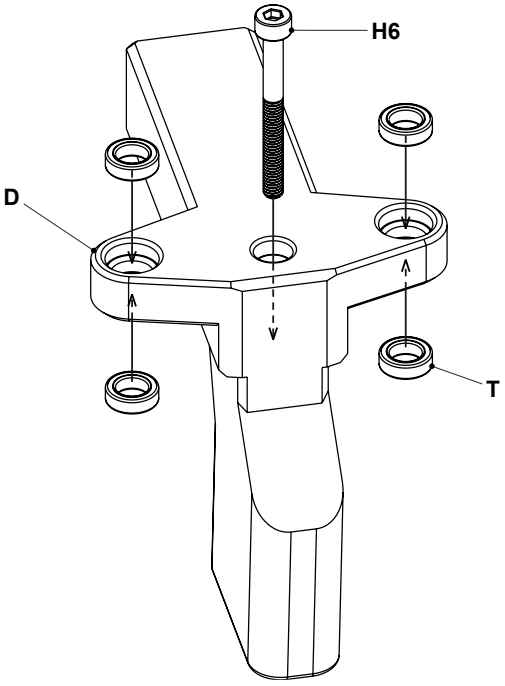
C **2x**
Gear axle

H4 **2x**
M3 × 20 mm

H8 **2x**
M3 nut

STEP 4

Prepare the trigger



Seat one small bearing on each face of both trigger pivots (four total). Insert the M3 x 30 screw into the central recessed hole.

PARTS ADDED

D **1x**
Trigger

T **4x**
Small bearing

H6 **1x**
M3 × 30 mm

STEP 5

Apply low-friction tape

PARTS ADDED

AP **Cut to fit**

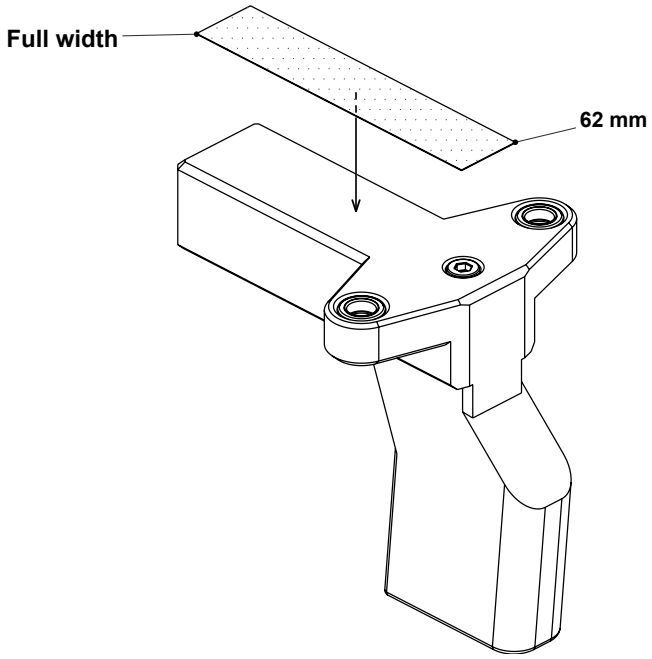
Low-friction tape

ALREADY ASSEMBLED

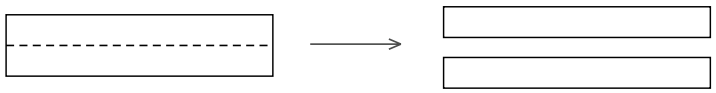
Prepared trigger

From Step 4

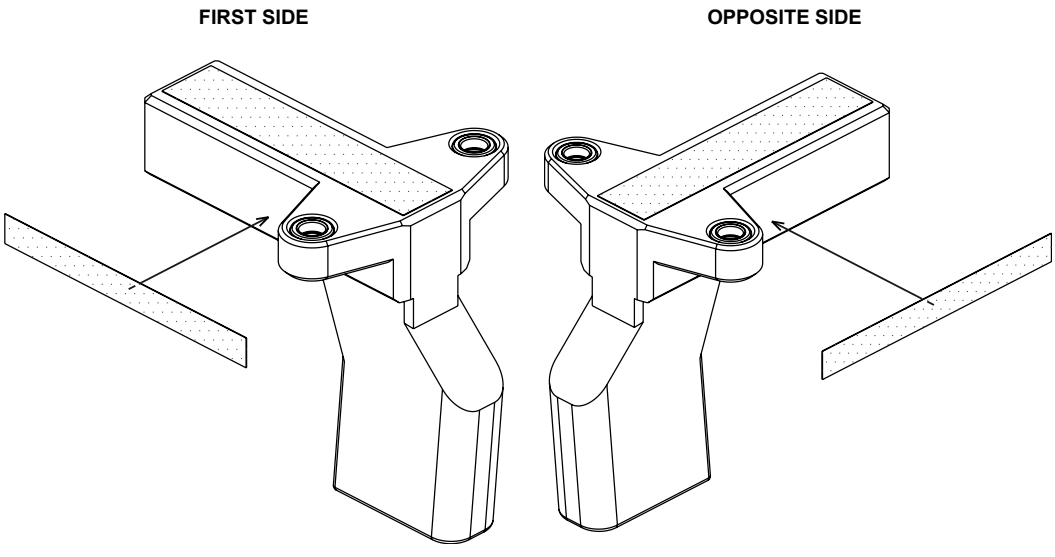
1. Top strip



2. Split the second strip lengthwise



One half-width strip on each side

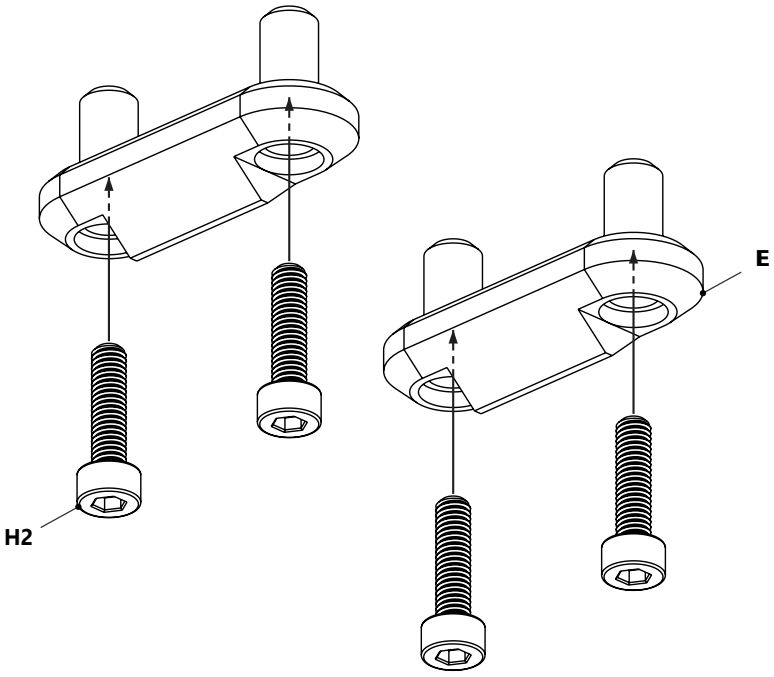


Cut two strips of tape, each 62 mm long. Apply one full-width strip along the trigger top, covering the recessed screw. Split the second strip lengthwise and apply one half-width strip to each side. Press flat and trim clear of the bearings and spring path.

Do not stack or overlap tape layers. Apply tape only to flat surfaces; do not bend around corners.

STEP 6

Prepare both linkages



Insert two M3 x 12 screws into each linkage from its underside. Seat the heads in the recessed holes. Prepare both linkages as shown, with the raised pivots ready to enter the trigger bearings in Step 7.

PARTS ADDED

E **2x**

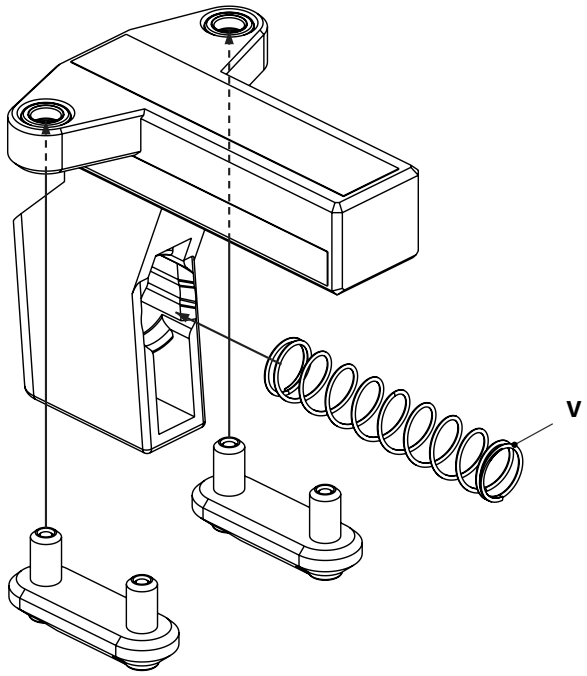
Linkage

H2 **4x**

M3 × 12 mm

STEP 7

Join the trigger and linkages



Bring both prepared linkages up from below into the trigger bearings. Slide the compression spring onto the trigger stem from the rear, along the axis shown. Keep the spring aligned as you bring the subassembly to the body.

PARTS ADDED

V **1x**

Compression spring

ALREADY ASSEMBLED

Prepared trigger

From Step 5

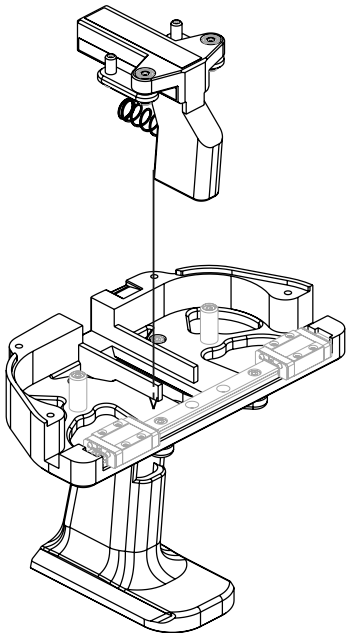
Prepared linkages

From Step 6

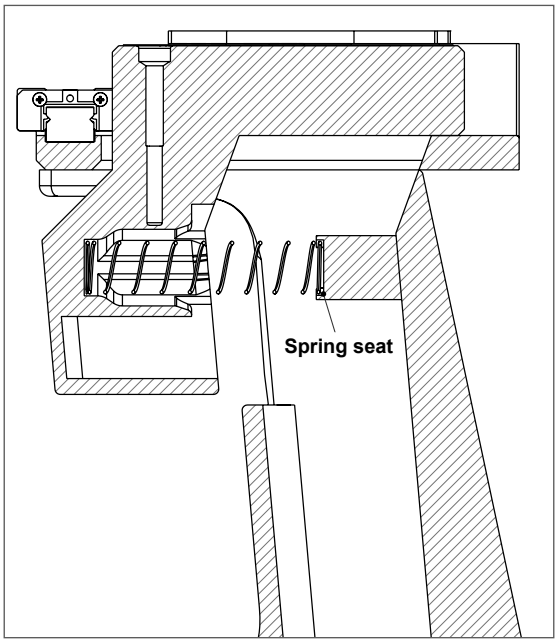
STEP 8

Seat the trigger in the body

LOWER INTO BODY



CENTRE SECTION / SEATED



Lower the trigger into the central guide. Seat the spring between the trigger and its stop in the grip. Ensure both linkages remain fully seated in their bearings.

Tweezers may be needed to position the spring correctly in its seat.

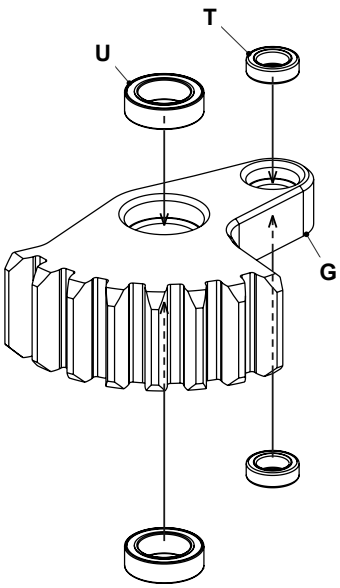
PARTS ADDED

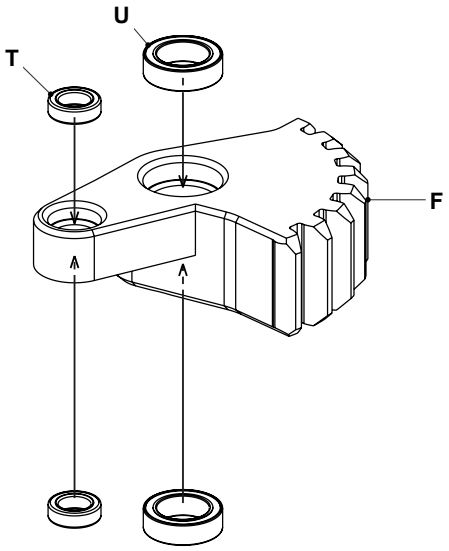
Body assembly
From Step 3

Trigger + linkages
From Step 7

STEP 9

Prepare the left and right gears





Fit two large MR128-2RS bearings to the main pivot of each gear and two small MR85-2RS bearings to its linkage pivot. Seat bearings squarely in their recesses.

Do not force the bearings into their recesses. If the fit is too tight, carefully file or adjust the gear holes until the bearings seat without excessive force.

PARTS ADDED

F **1x**
Left gear

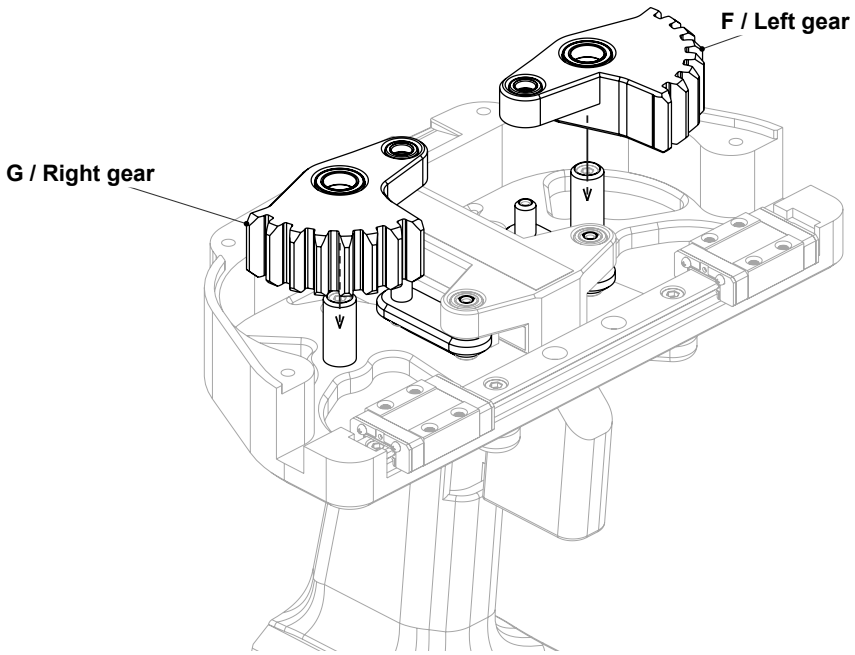
G **1x**
Right gear

U **4x**
Large bearing

T **4x**
Small bearing

STEP 10

Install and engage the gears



Place the prepared gears over the two printed axles. Engage the linkage pivots and align the teeth toward the finger-holder side, as shown. Move the trigger gently to check both sides travel together.

PARTS ADDED

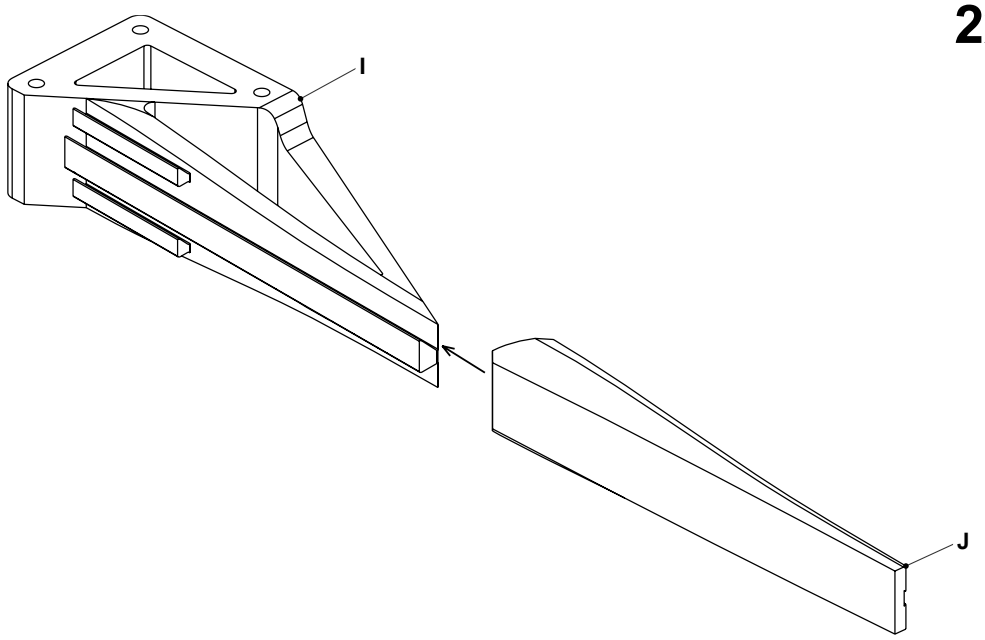
Body + trigger
From Step 8

Prepared gears
From Step 9

STEP 11

Fit the TPU finger pads

2x



Slide each TPU pad onto its backbone from the finger tip (front). Keep it aligned with the backbone and push it backward along the full length until seated. Build a mirrored pair with the padded faces facing one another.

The pad slides on from the front; do not push it on sideways.

PARTS ADDED

I **2x**
Finger backbone

J **2x**
Finger pad

STEP 12

Apply the grip tape

PARTS ADDED

AN **Cut to fit**

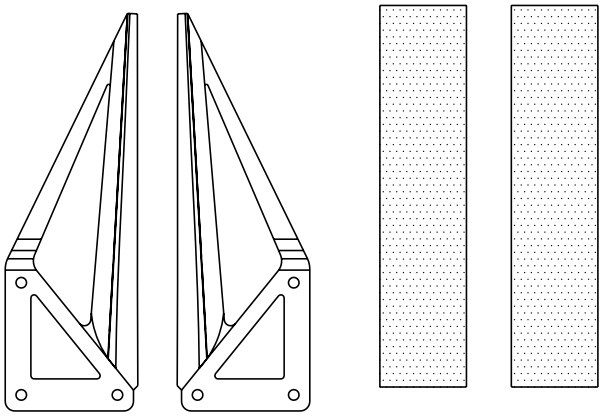
Grip tape

ALREADY ASSEMBLED

Finger pair

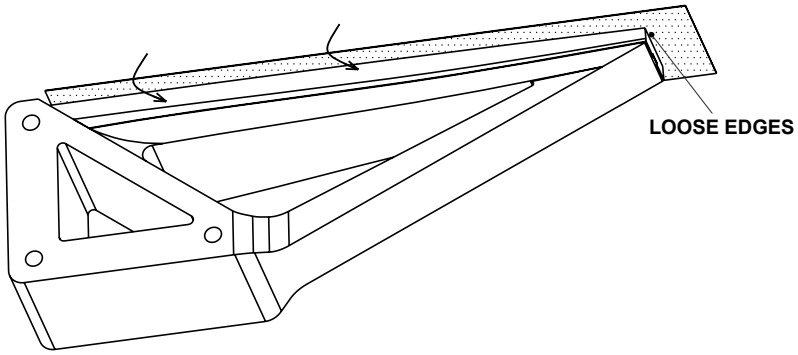
From Step 11

1. Ready for tape



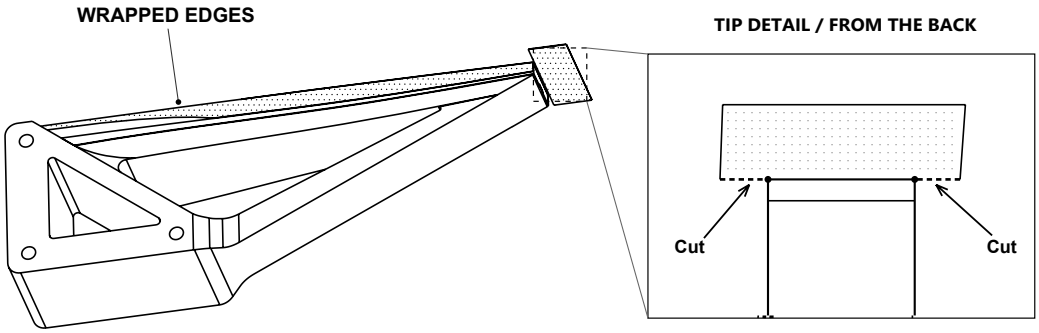
Start with both padded fingers from Step 11. Cut one tape strip for each, leaving enough to wrap the long edges and tip.

2. Apply the tape / leave the tip long



Center the strip along the TPU contact face. Leave tape overhanging both long edges and the finger tip.

3. Free the end tab / wrap the long edges



Cut inward from each tape edge to the tip corner (detail). Leave the center uncut, then smooth the long edges around the pad.

Continue with the enlarged tip views on the next page. Cut only the tape; keep the TPU pad and printed backbone intact.

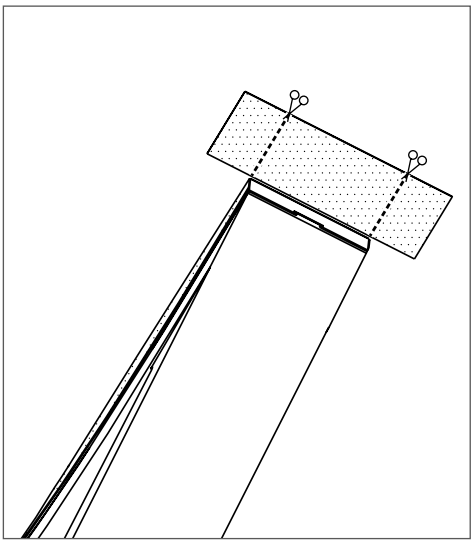
STEP 12 CONT.

Finish the grip tape

CONTINUE WITH

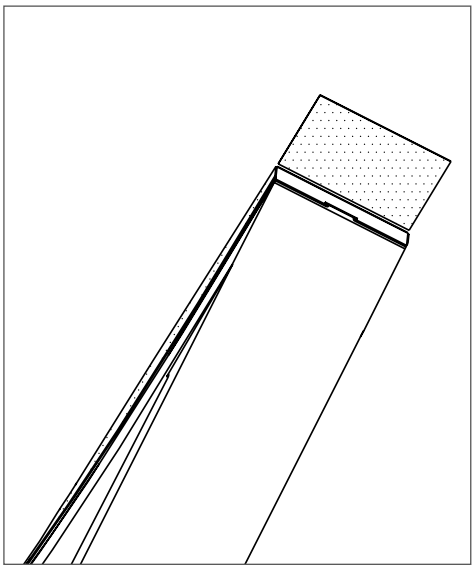
Same fingers and grip tape from the previous page.

4. Wide tab / trim the sides



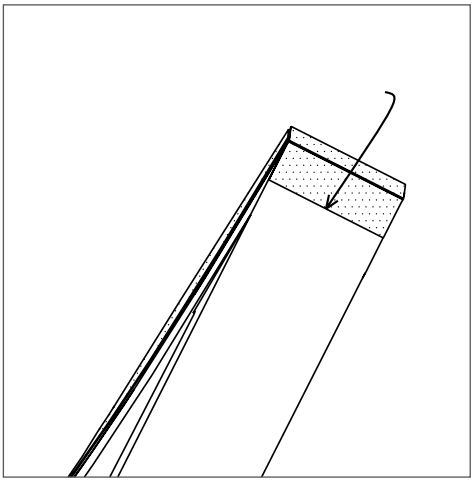
View the flat back of the tip. Trim the tab sides along the dashed lines, aligned with the two edges of the white backbone.

5. Tab trimmed to width



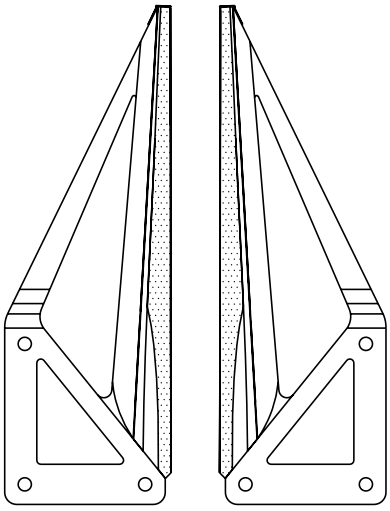
The tab now matches the flat back of the finger. Keep this center flap long enough to fold over the tip.

6. Fold flat over the tip



Fold the flap over the tip and press it flat against the back of the finger.

7. Finished pair



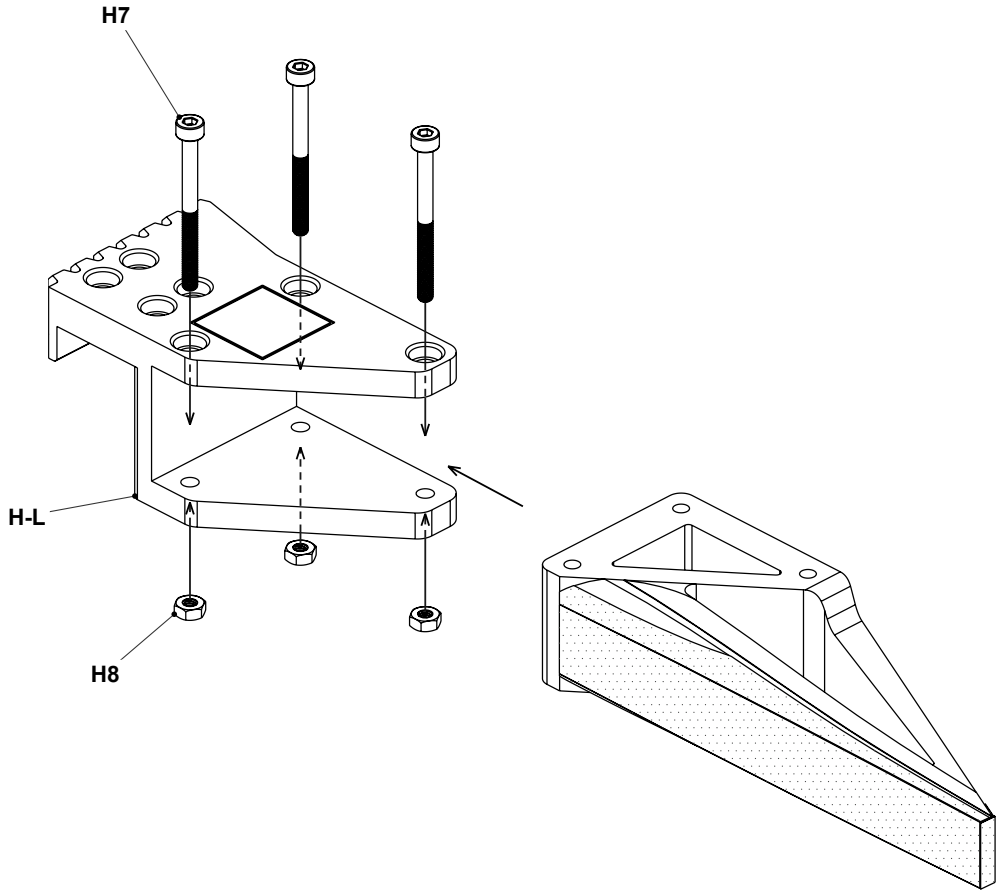
Repeat for the other finger.

Press all tape edges and the folded tip firmly into place.
Trim any loose material clear of the mechanism.

STEP 13

Fit a finger to its holder

2x



Slide one completed finger into the left holder (H-L), shown, from the finger-tip side along the arrow. Align the three holes and fit three M3 × 35 mm fasteners and three nuts. Attach one ArUco tag in the square recess. Repeat with the right holder (H-R) and the other finger.

Build one H-L and one H-R assembly. See tag sizing and IDs on page 15.

PARTS ADDED

H-L **1x**

Left finger holder

H-R **1x**

Right finger holder

H7 **6x**

M3 × 35 mm

H8 **6x**

M3 nut

AM **2x**

ArUco tags

ALREADY ASSEMBLED

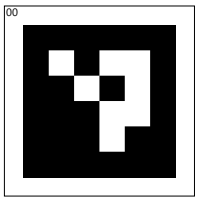
Finger pair

From Step 12

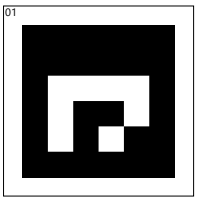
Fit the tags

Each unit uses two tags, one within each recessed border on the finger holders. Print the supplied ArUco tag sheet at **Actual size / 100%**, with Fit and Shrink disabled. Verify the 50 mm scale check before cutting.

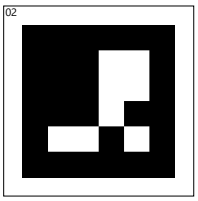
SETUP	TAG IDS
Each UMI-Live	Two distinct IDs
Left / right assignment	Configurable in software



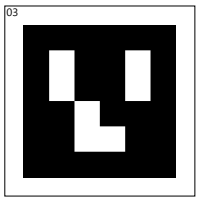
ID 0



ID 1



ID 2



ID 3

TAG PATTERNS SHOWN FOR IDENTIFICATION; PRINT FROM THE SUPPLIED TAG SHEET.

Size and placement

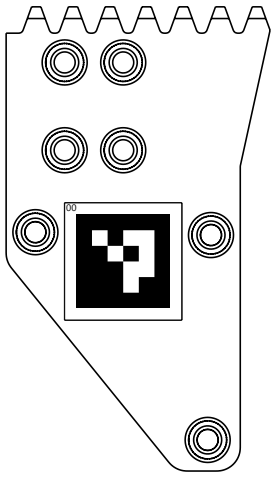
Dictionary: DICT_4X4_50.

Black marker: 16 x 16 mm, including its black border.

Cut size: 20 x 20 mm; retain the 2 mm white margin.

Attachment: tape or glue.

Keep the printed face flat, clean and visible.



TAG WITHIN RECESSED BORDER

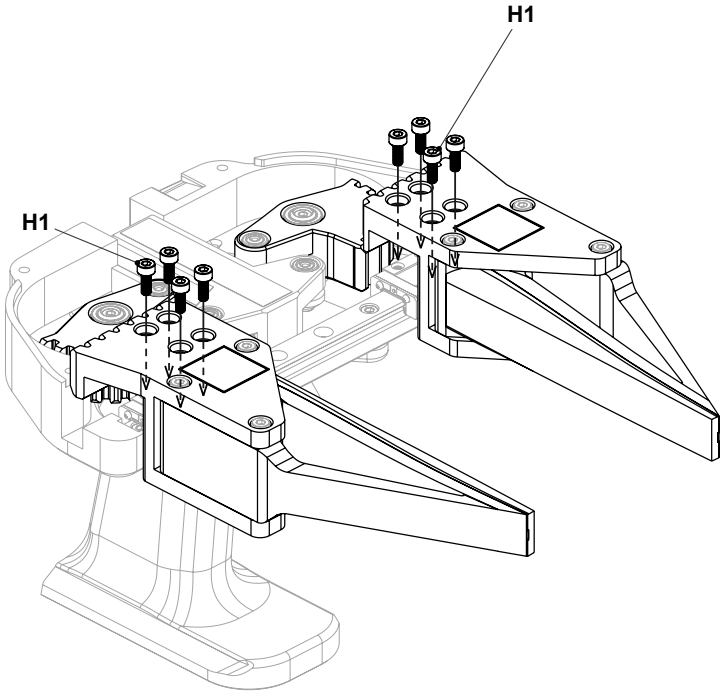
Tag orientation does not matter. The tag IDs for each UMI-Live gripper can be configured in software.

Fit each tag within the recessed border without covering the white margin. Keep the printed face flat, clean and visible.

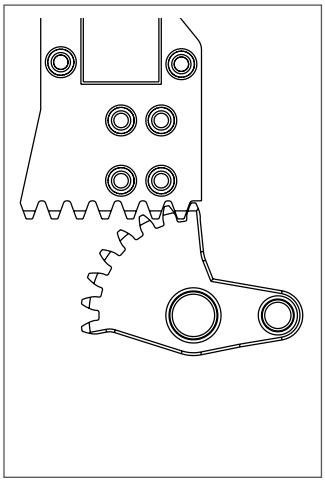
Print from ArUco_tags_0-3_all_16mm_US_Letter.pdf (IDs 0-3).
Do not cut tags from the enlarged diagrams in this manual.

STEP 14

Attach both finger assemblies



TOP VIEW / LEFT SIDE



F meshes with H-L

Mesh the left holder (H-L) with the left gear (F), and the right holder (H-R) with the right gear (G). Align each holder over its carriage and secure it with four M3 × 8 mm fasteners. Keep the padded faces facing each other and check that both fingers move symmetrically.

PARTS ADDED

H1 **8x**

M3 × 8 mm

ALREADY ASSEMBLED

Body + gears

From Step 10

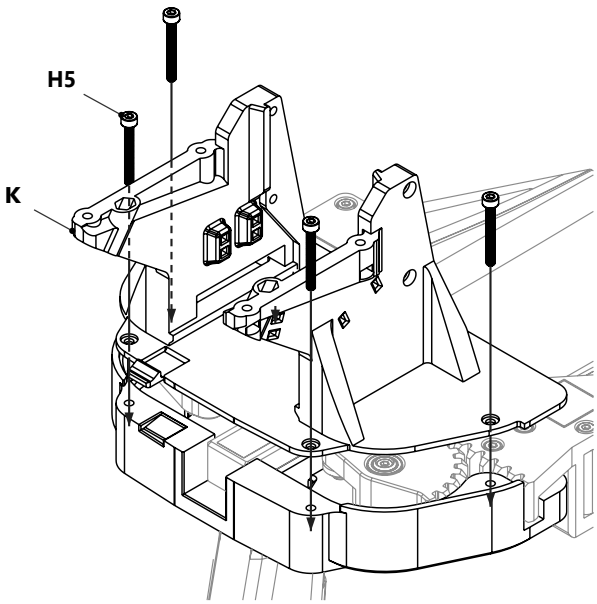
Finger assemblies

From Step 13

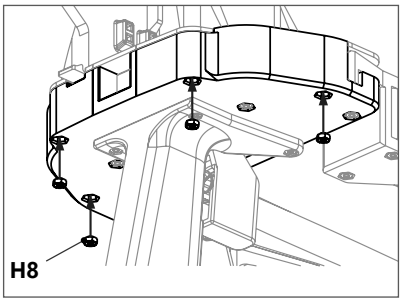
STEP 15

Close the body

BODY CLOSURE / CROPPED VIEW



NUT POCKETS / UNDERSIDE



Lower the body top over the completed mechanism. Fit four M3 x 25 screws and nuts. Tighten gradually, checking trigger return and free carriage motion as you go.

PARTS ADDED

K **1x**

Body top

H5 **4x**

M3 × 25 mm

H8 **4x**

M3 nut

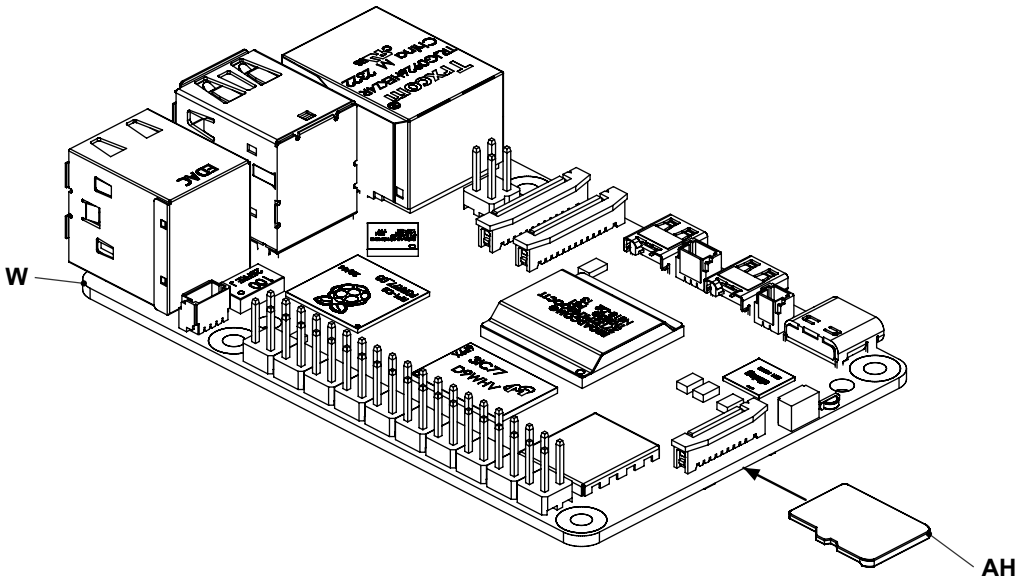
ALREADY ASSEMBLED

Body + fingers

From Step 14

STEP 16

Prepare the Pi



Insert the microSD card into the underside slot, contact end first, with the gold contacts facing the board and the label facing away.

Slide it in gently; do not force it. Prepare the microSD card using the UMI-Live software setup guide before first use. Software setup can be completed after assembly.

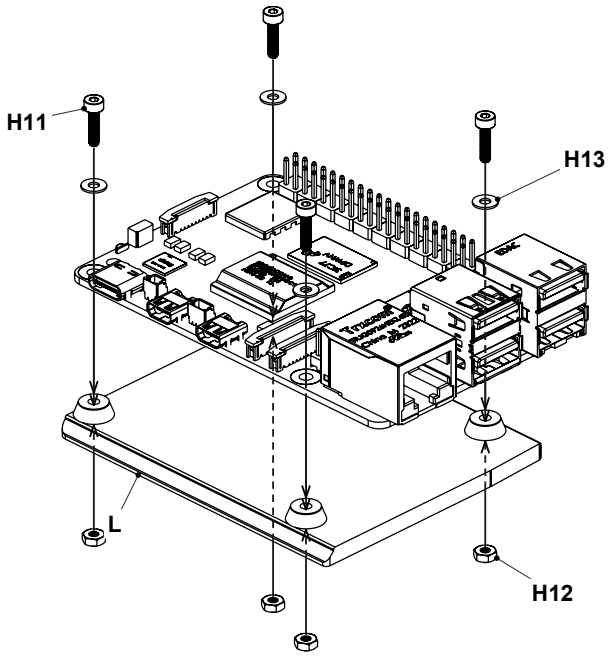
PARTS ADDED

W **1x**
Raspberry Pi 5

AH **1x**
microSD card

STEP 17

Mount the Pi on its slider



Place the Pi on its slider and align the four mounting holes. Fit four M2 x 8 screws, washers and nuts. Tighten just enough to hold the PCB without bending it.

Place each washer directly under its screw head, against the PCB.

PARTS ADDED

L **1x**
Pi slider

H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

ALREADY ASSEMBLED

Prepared Pi
From Step 16

LCD + button pin map

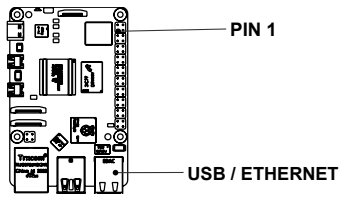
GPIO numbers use **BCM numbering**; pins use physical numbering. With the Pi **component side up**, orient the USB/Ethernet ports at the bottom and the header on the right. Pin 1 is the header's upper-left pin, farthest from those ports. Keep power disconnected.

LCD	PI SIGNAL	PIN
VCC	3.3V	1
GND	Ground	6
DIN	GPIO10 / MOSI	19
CLK	GPIO11 / SCLK	23
CS	GPIO8 / CE0	24
DC	GPIO25	22
RST	GPIO27	13
BL	GPIO18	12

Button connections

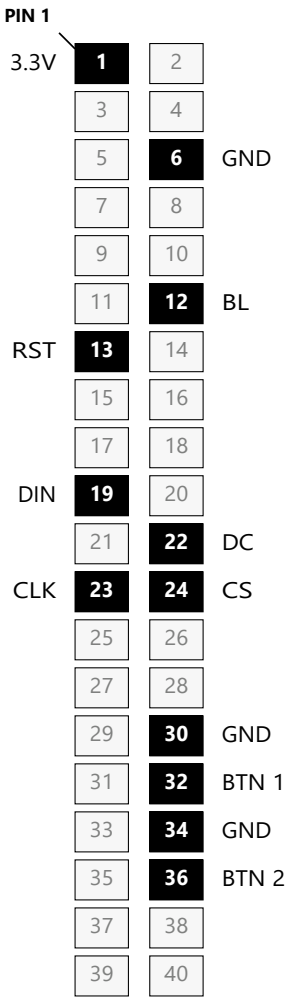
BUTTON	SIGNAL	GROUND
BTN 1 Near fingers	GPIO12 / pin 32	Pin 30
BTN 2 Near LCD	GPIO16 / pin 36	Pin 34

COMPONENT SIDE UP



40-PIN HEADER

COMPONENT SIDE / PHYSICAL PINS



USB / ETHERNET END OF BOARD

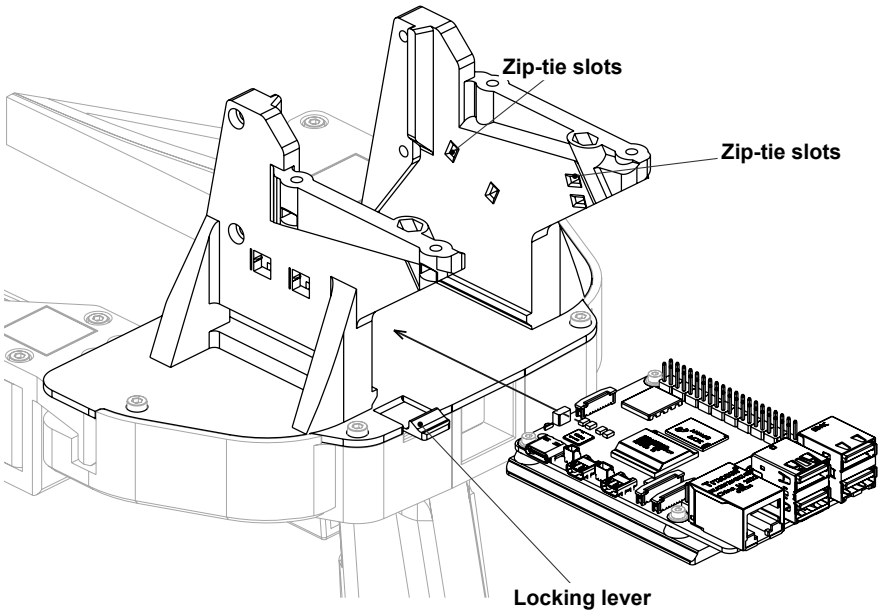
> **!** GPIO12 is physical pin 32. Physical pin 12 is LCD backlight GPIO18.

Buttons: use female-to-female leads. One terminal goes to the specified GPIO; the other goes to its paired ground. All four leads occupy the even-numbered row at pins 30, 32, 34 and 36.

LCD: follow the signal labels, rather than cable colors. Use Pi 3.3V for VCC in this build. The camera uses USB 3.0; Airy uses Ethernet and separate power.

STEP 18

Slide the Pi into the body



Slide the Pi and slider forward into the body top guides until the lever clicks and locks. Thread optional zip ties through the marked slots for cable management; leave them loose until wiring is complete.

PARTS ADDED

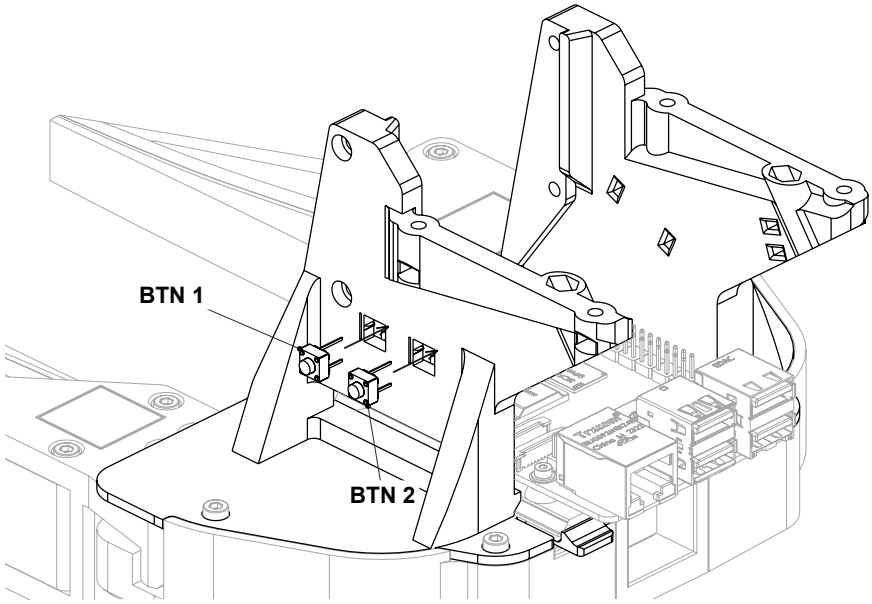
Body assembly
From Step 15

Pi + slider
From Step 17

2 optional
Zip ties

STEP 19

Install the two buttons



Seat the buttons in the side openings. Connect the button nearest the fingers to GPIO12 and GND (pins 32 and 30). Connect the button nearest the LCD to GPIO16 and GND (pins 36 and 34).

Use four female-to-female leads; connect both ends using page 18. Push each socket fully onto its terminal and gently tug-check the connection.

PARTS ADDED

X **2x**
Tactile button

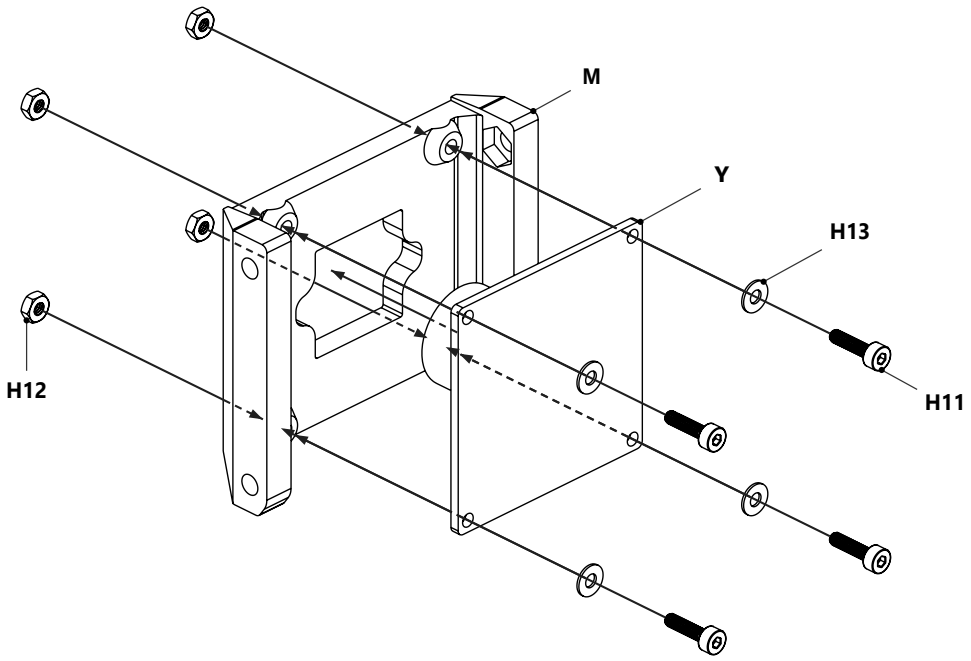
AL **4x**
Button leads

ALREADY ASSEMBLED

Body + Pi
From Step 18

STEP 20

Mount the camera board



Bring the camera board forward from behind the mount, with the lens facing through the opening. Fit four M2 x 8 screws, washers and nuts. Connect the 4-pin JST PH end of the USB camera cable to the camera board.

Washers go directly under the screw heads. Keep the lens and PCB clear.

PARTS ADDED

M **1x**
Camera mount

Y **1x**
SVPRO USB camera

From camera **1x**
USB Camera Cable

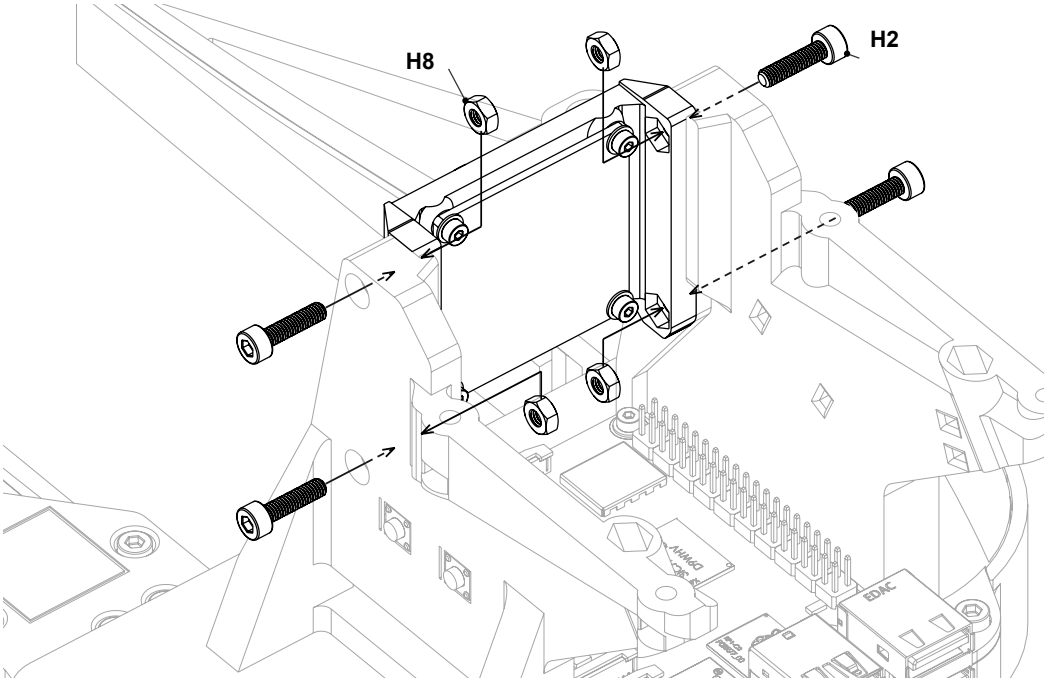
H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

STEP 21

Attach and connect the camera



Secure the camera mount to the front of the body top with four M3 x 12 screws and nuts. Plug the USB camera cable into a blue USB 3.0 Type-A port on the Raspberry Pi.

PARTS ADDED

H2 **4x**
M3 × 12 mm

H8 **4x**
M3 nut

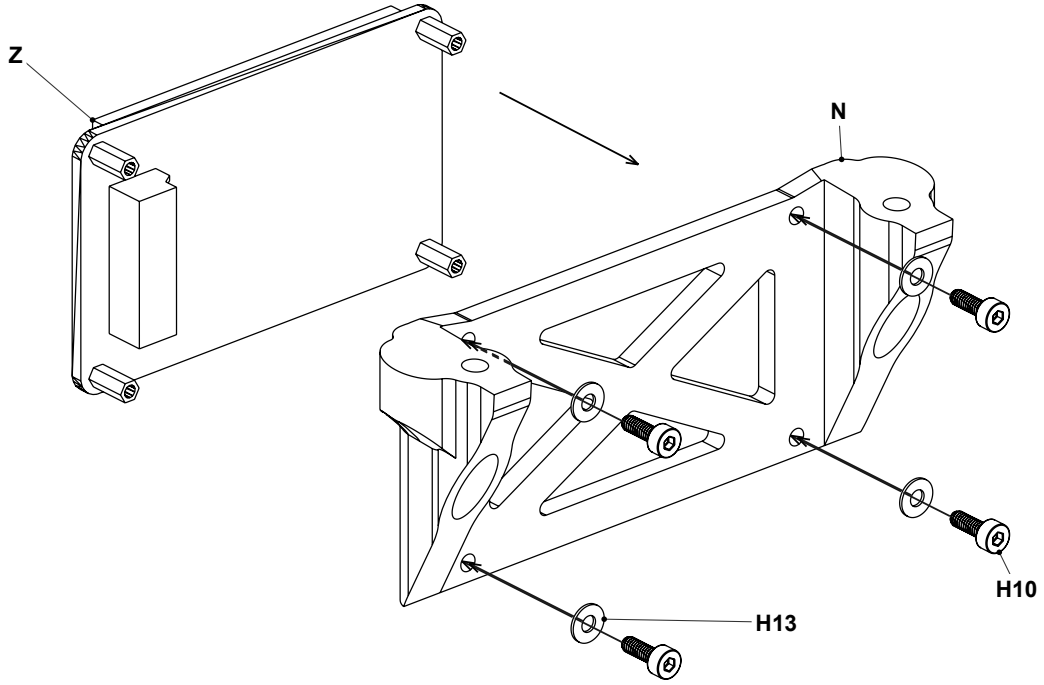
ALREADY ASSEMBLED

Body + buttons
From Step 19

Camera assembly
From Step 20

STEP 22

Fit the LCD to its mount



Place the LCD in its printed mount with the display facing outward. Fit four M2 x 6 screws and washers at the mounting holes. Keep the display cable and connector clear of the frame.

The four washers sit under the M2 screw heads.

PARTS ADDED

N **1x**
LCD mount

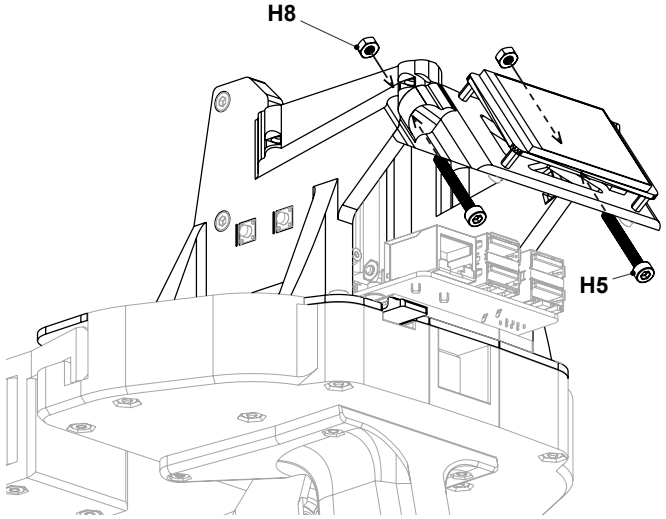
Z **1x**
Waveshare LCD

H10 **4x**
M2 × 6 mm

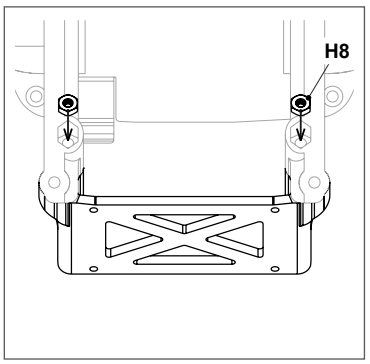
H13 **4x**
M2 washer

STEP 23

Attach and connect the LCD

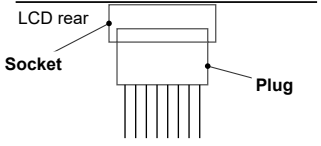


TOP VIEW / NUT SLOTS



LCD omitted for clarity

LCD PLUG / SEATED (SCHEMATIC)



Seat plug against socket

Mount the LCD assembly at the rear of the body using two M3 x 25 screws and nuts. Connect the eight-wire LCD cable to the LCD and Pi header using the pin map on page 18. Recheck every signal before applying power.

Ensure that the LCD cable is fully pushed into the LCD display.

PARTS ADDED

H5 **2x**
M3 × 25 mm

H8 **2x**
M3 nut

From LCD **1x**
LCD cable

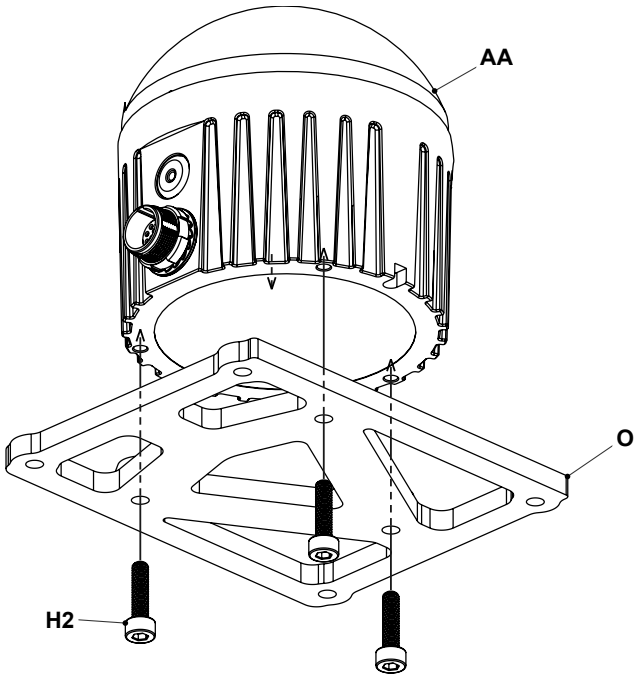
ALREADY ASSEMBLED

Body + camera
From Step 21

LCD assembly
From Step 22

STEP 24

Fasten Airy to the LiDAR mount



Align the three mounting holes in the Airy base with the LiDAR mount. Use three M3 x 12 screws through the plate into the LiDAR. Seat the unit evenly and avoid bottoming the screws.

Keep the LiDAR optical dome and ventilation clear.

PARTS ADDED

O **1x**

LiDAR mount

AA **1x**

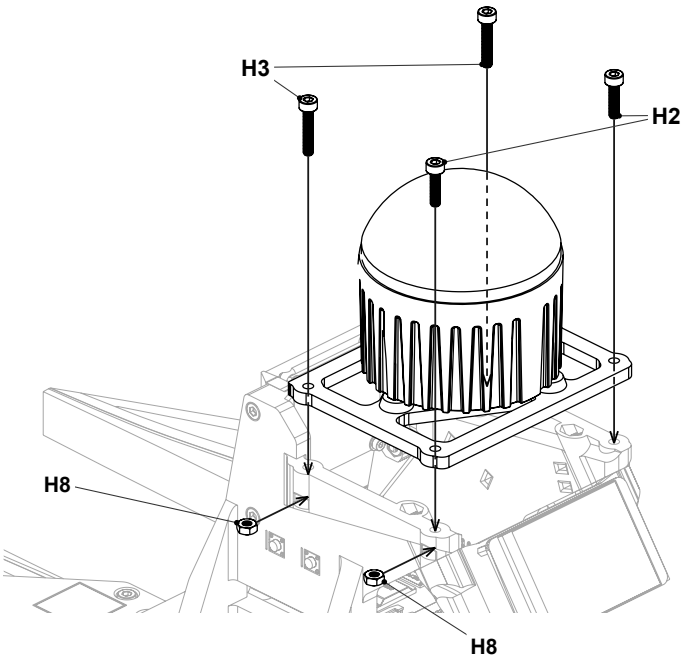
RoboSense Airy LiDAR

H2 **3x**

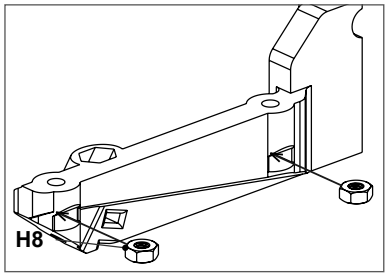
M3 × 12 mm

STEP 25

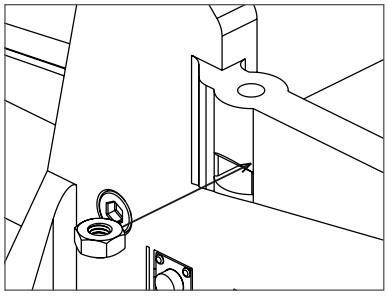
Install the LiDAR assembly



OPPOSITE SIDE / CUTAWAY



VISIBLE SIDE / SLOT DETAIL



Slide two M3 nuts inward from each outer side into the four post slots. Place the LiDAR mount on the supports. Fit M3 x 16 screws at the front holes (camera side) and M3 x 12 screws at the rear (LCD side).

Connect the Airy cable to the LiDAR by hand screwing it in. Then plug one end of the Ethernet cable into the Airy cable and the other end into the Raspberry Pi.

PARTS ADDED

H2 **2x**

M3 × 12 mm

H3 **2x**

M3 × 16 mm

H8 **4x**

M3 nut

AK **1x**

Ethernet cable

From LiDAR **1x**

Interface cable

ALREADY ASSEMBLED

Body

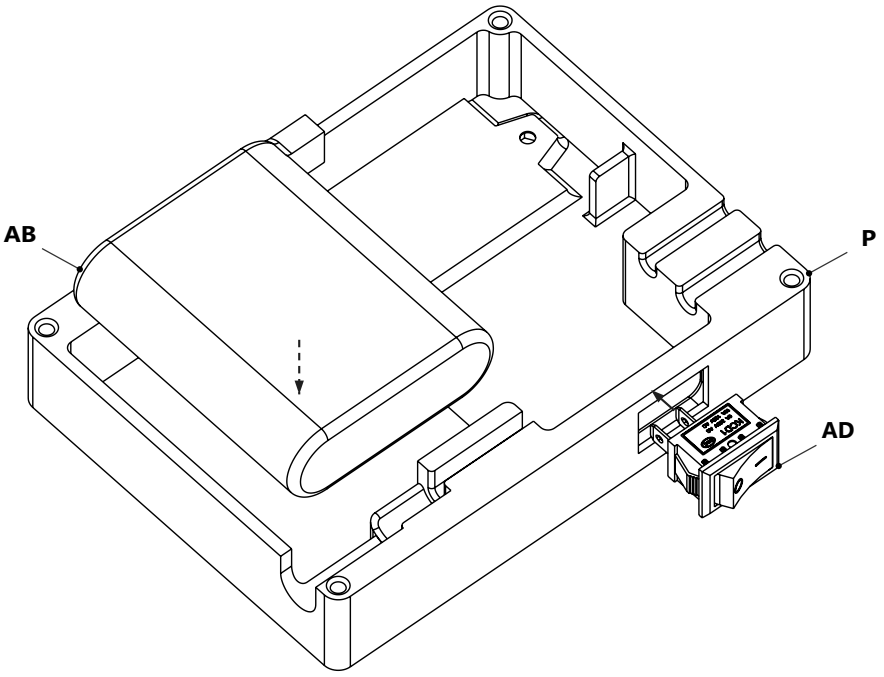
From Step 23

LiDAR assembly

From Step 24

STEP 26

Fit the battery and switch



With the battery disconnected and switch OFF, seat the battery in the large compartment and clip the rocker switch into the side opening. Route its leads toward the connector compartment.

PARTS ADDED

P **1x**

Battery pack bottom

AB **1x**

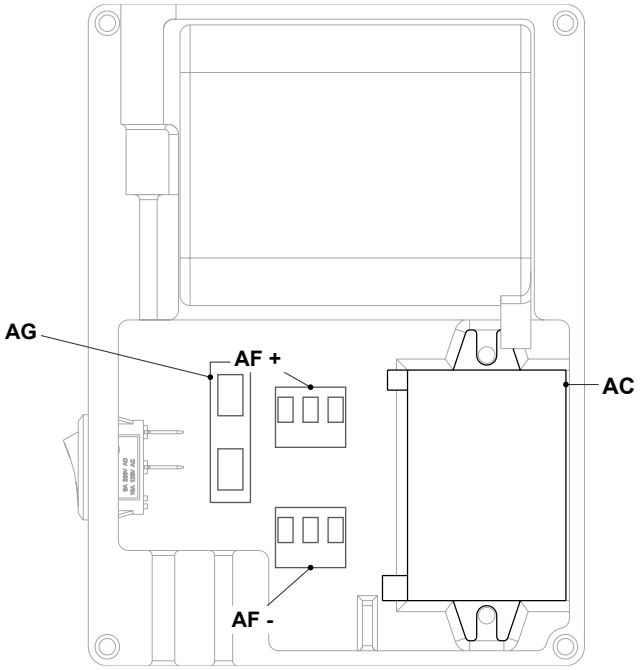
Battery

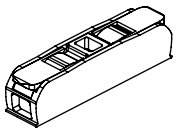
AD **1x**

Power switch

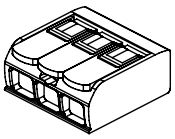
STEP 27

Prepare the power harness

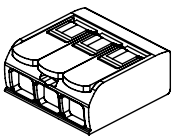




AG / Inline



AF / Positive (+)



AF / Negative (-)

Placement shown; circuit on page 24.

Use one 221-2401 between battery positive and the switch. Use one 221-413 for switched positive and one for negative. Split both branches to the converter input and Airy power lead. Follow the battery circuit diagram.

Keep the battery disconnected. Strip to each connector's marked length; insert fully, close the levers and tug-test.

PARTS ADDED

AC **1x**

DC-DC converter

From battery **1x**

DC barrel lead

AF **2x**

WAGO 221-413

AG **1x**

WAGO 221-2401

ALREADY ASSEMBLED

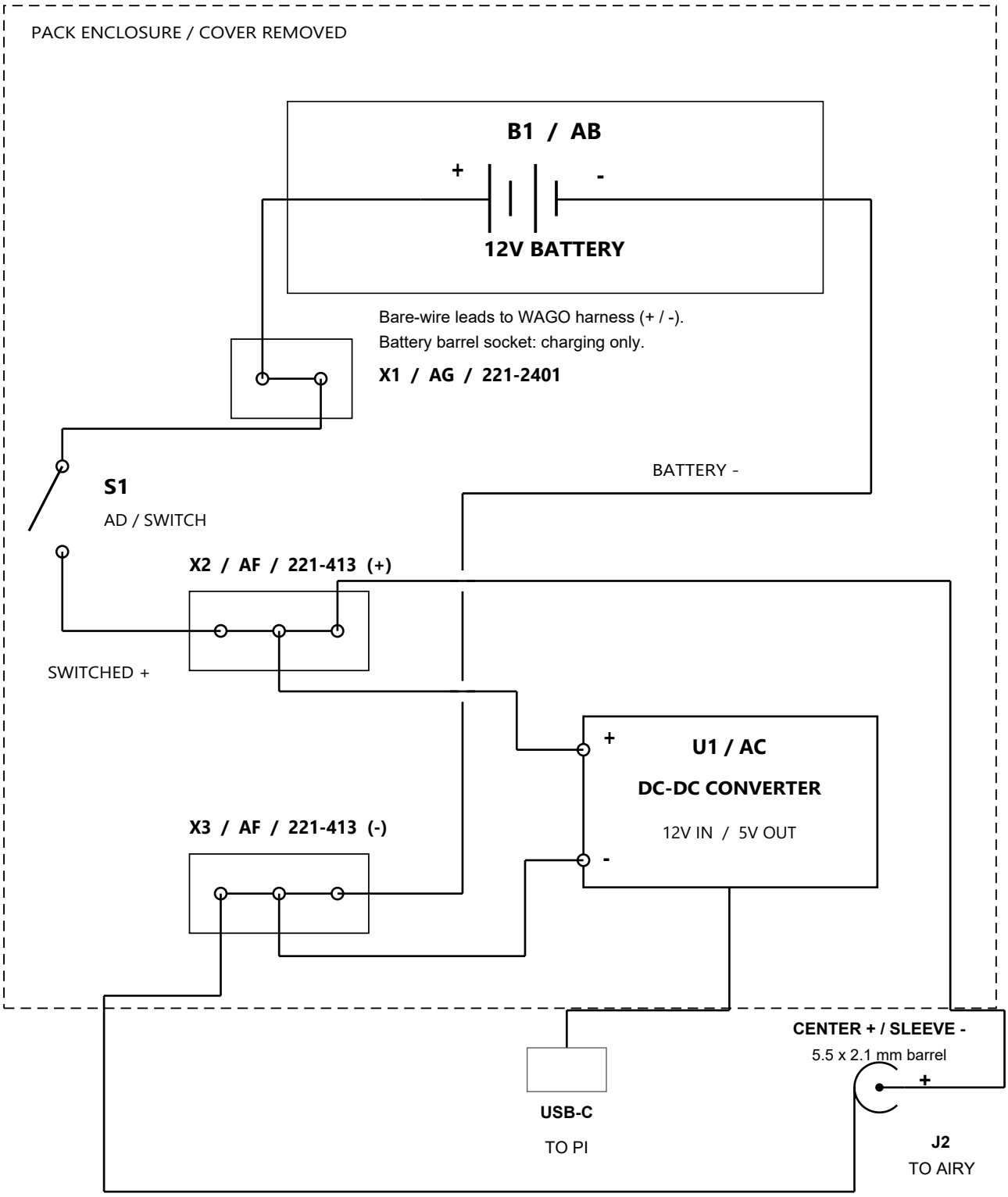
Open battery pack

From Step 26

Battery pack circuit

Top view of the open pack: battery above, switch on the left, converter at lower right, and WAGO connectors in the remaining compartment. Wire lengths and routing are illustrative.

For the Airy barrel-output test: set the multimeter to DC voltage. Place red on the barrel center and black on the outer sleeve; the reading must be positive and around 12 V.

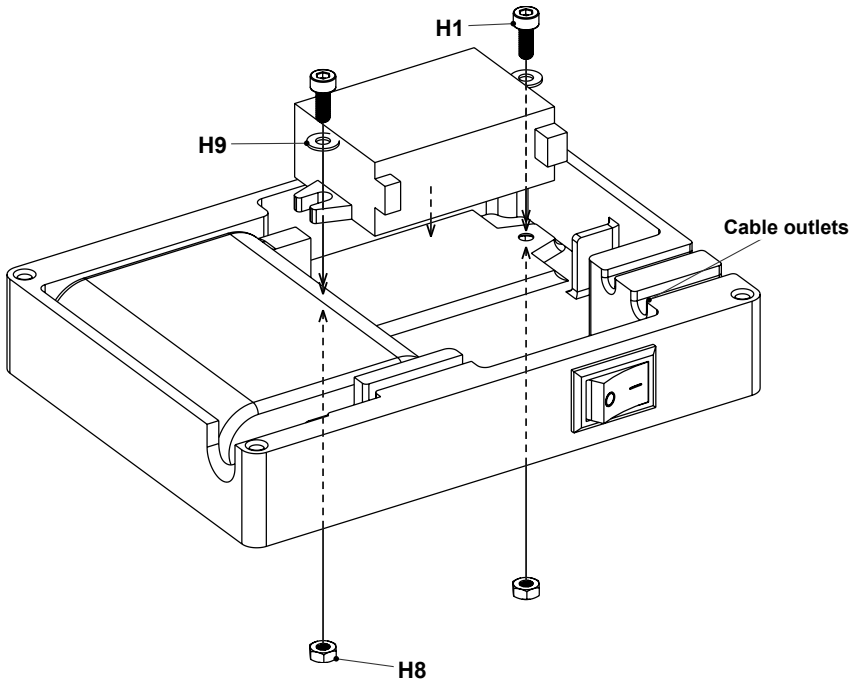


All electrical conductors use solid lines. A filled dot means joined; a gap at a crossing means the wires are not connected. WAGO terminals inside each box are internally common. USB-C is shown as a complete cable.

Output test: leave the Pi and Airy unplugged. Connect the battery, switch ON, and verify center-positive polarity and around 12 V at the Airy barrel output. Then switch OFF and disconnect the battery before securing the converter and lid.

STEP 28

Secure the converter



Fit the converter beside the battery using two M3 x 8 screws, two washers and two nuts. Place the connectors in the remaining compartment. Route the USB-C and LiDAR power leads through the case outlets.

Fit a washer under each screw head. The two nuts sit beneath the case.

PARTS ADDED

H1 **2x**
M3 × 8 mm

H8 **2x**
M3 nut

H9 **2x**
M3 washer

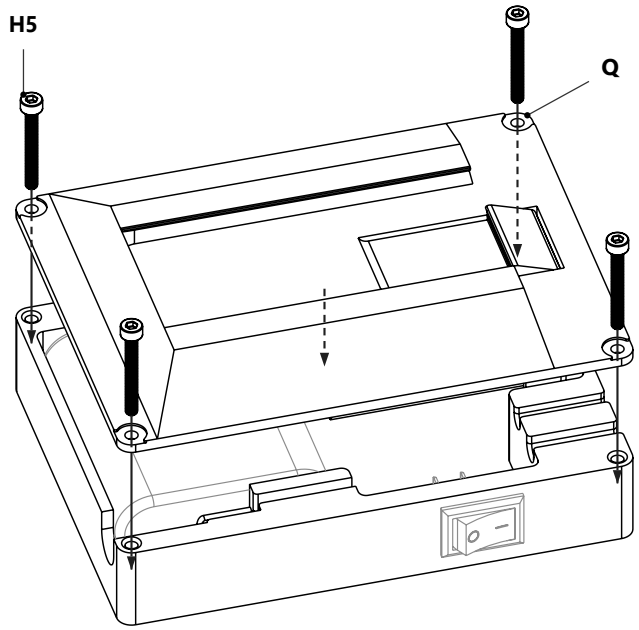
ALREADY ASSEMBLED

Pack + harness
From Step 27

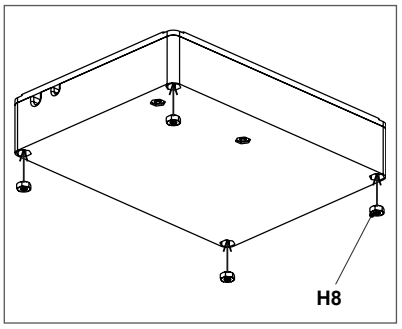
STEP 29

Close the battery pack

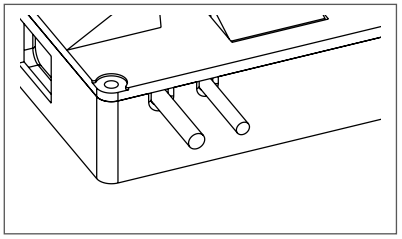
COVER FITTING



NUT POCKETS / UNDERSIDE



CABLE EXITS / LID SEATED



Keep both cables in their channels.

After the output test on page 24 passes, keep the switch OFF and reconnect the battery. Place the lid without pinching wires. Fit four M3 x 25 screws and nuts. Check that the switch is accessible and both output cables are held securely.

PARTS ADDED

Q **1x**
Battery pack top

H5 **4x**
M3 × 25 mm

H8 **4x**
M3 nut

ALREADY ASSEMBLED

Battery pack
From Step 28

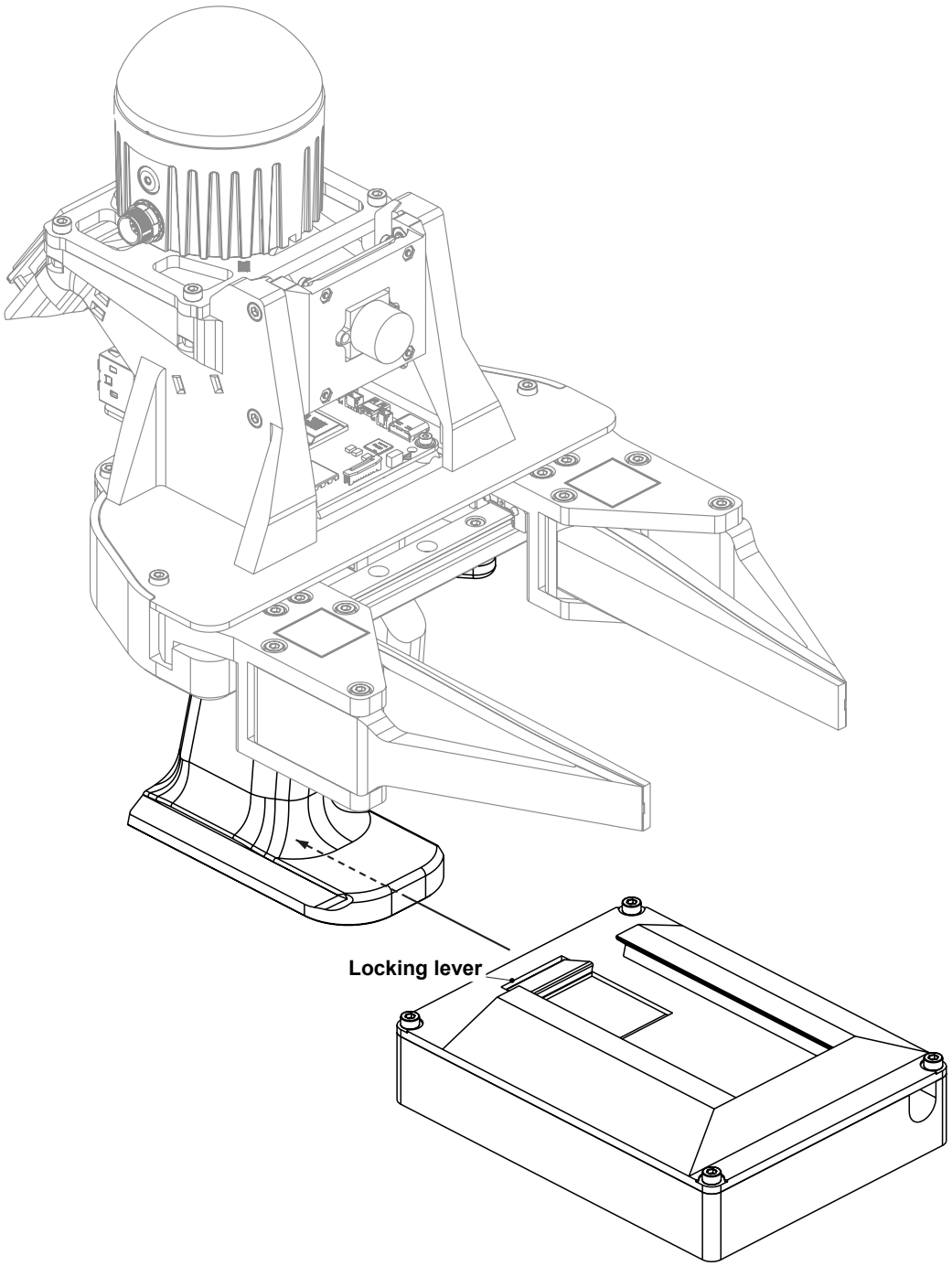
STEP 30

Slide on the battery pack

PARTS ADDED

UMI-Live assembly
From Step 25

Battery pack
From Step 29

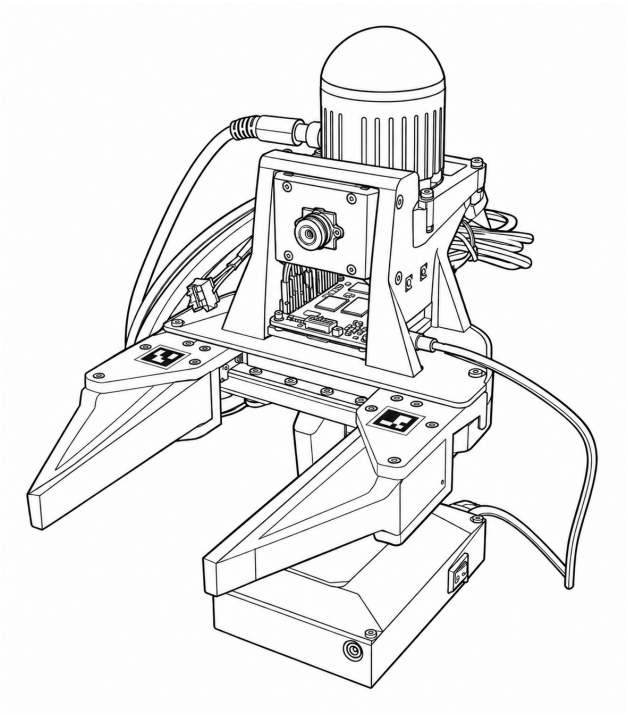


Align the battery-pack rails with the base of the grip.
Slide the pack until it clicks and locks into place.

If the battery pack lever is unable to be pushed down, open the battery pack and ensure that the components inside are not obstructing its motion.

STEP 31

Connect the finished hardware



CABLE ROUTING REFERENCE

Illustrated from the UMI-Live repository photo.

With the switch OFF, connect the pack USB-C output to the Pi power input and its DC barrel output to the Airy power input. Confirm Airy Ethernet goes to the Pi RJ45 port and the camera goes to USB 3.0.

Complete the final checks below before switching ON.

PARTS ADDED

Completed hardware
From Step 30

CAMERA
USB 3.0 to Pi

AIRY DATA
Ethernet to Pi

PI POWER
Pack USB-C

AIRY POWER
Pack DC barrel to original cable

Final check

- ☐ Trigger returns under spring force; both fingers move smoothly together.
- ☐ Fasteners are seated; bearings and carriages remain free.
- ☐ Tags are flat, visible and have distinct IDs.
- ☐ Buttons use pins 32/30 and 36/34. LCD follows the pin map.
- ☐ Camera connects to USB 3.0; Airy connects to Ethernet and its original power cable.
- ☐ No bare conductors, trapped wires or cable loops in the moving mechanism.
- ☐ Pack output test on page 24 passed; both assemblies are fully seated.

Charge the battery

1. Switch OFF. Disconnect all connections to the battery, including both bare-wire leads from the WAGO harness.
2. Plug the supplied charger directly into the battery's charging barrel socket.
3. Unplug the charger before reconnecting the battery to the harness.

Do not charge and use the battery at the same time.